

LESSON 10.4

SOLVING ONE-STEP INEQUALITIES USING MULTIPLICATION AND DIVISION



LESSON INTRODUCTION

MA.7.AR.2.1 Write and solve one-step inequalities in one variable within a mathematical context and represent solutions algebraically or graphically.

LEARNING TARGETS

- I can solve one-step inequalities in one variable using multiplication or division.
- I can relate properties of inequality to solving one-step inequalities algebraically.

BENCHMARK COHERENCE

BUILDING ON

MA.6.AR.1.2 Translate a real-world written description into an algebraic inequality in the form of $x > a$, $x < a$, $x \geq a$, or $x \leq a$. Represent the inequality on a number line.

MA.6.AR.2.1 Given an equation or inequality and a specified set of integer values, determine which values make the equation or inequality true or false.

WORKING TOWARD

MA.8.AR.2.2 Solve two-step linear inequalities in one variable and represent solutions algebraically and graphically.

ESSENTIAL QUESTIONS

- How do I use properties of inequalities to solve one-step multiplication and division inequalities?

LESSON NARRATIVE

In this lesson, students first explore the effect multiplying by -1 has on an inequality. Students also explore properties of inequality to get an understanding of the process of solving inequalities. Then students use properties of inequality to solve one-step multiplication and division inequalities. Students also graph the solution sets and test values to prove their solutions are true.

COMPONENT SUMMARY

10.4.1 WARM-UP (~5 MIN.)

Determine whether a value is part of the solution set

10.4.3 BRING IT TOGETHER (5 MIN.)

Summarize the effect of a negative coefficient on an inequality

10.4.5 GUIDED PRACTICE (10 MIN.)

Practice solving and graphing one-step inequalities with multiplication and division

10.4.7 WRAP-UP (~5 MIN.)

Compare the process of solving inequalities with positive and negative coefficients

10.4.2 COLLABORATION (10 MIN.)

Explore the effect of a negative coefficient on an inequality

10.4.4 GUIDED INSTRUCTION (10 MIN.)

Solve one-step inequalities with multiplication and division

10.4.6 YOUR TURN! (10 MIN.)

Independent practice solving and graphing one-step inequalities with multiplication and division

RESOURCES

GLOSSARY

Key Terms:

- Multiplication property of inequality
- Division property of inequality

Additional Terms:

- Inequality
- Reciprocal
- Solution set
- Opposite

MATERIALS

- Calculator
- (Optional) Number line templates
- (Optional) Chart paper & markers

ADDITIONAL PREPARATION

- Consider creating an anchor chart with instructions, examples, and visuals for solving one-step inequalities with multiplication and division.

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may not realize that $-x$ has a coefficient of -1 .
- Students may struggle reading and interpreting inequalities with the variable on the right side.

THINGS TO CONSIDER

- Consider reviewing the term “reciprocal” and sharing examples of multiplying by the reciprocal.
- Consider asking students to practice reading inequalities from right to left and left to right.
- Consider having no-calculator questions where students turn their calculator over. Teachers should look for and consider opportunities for students to improve their fluency with rational numbers.
- If time is an issue, then consider assigning 10.4.6 Your Turn! for homework.

LITERACY AND LANGUAGE ROUTINES

Think-Pair-Share

Use 10.4.2 Collaboration question 1 as an opportunity for students to initially try the problems independently and then discuss their strategies and answers with their partner. These discussions will help students gain confidence and understanding.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.4.1 Discussion and Reflection

In 10.4.5 Guided Practice, students discuss errors they identified and corrected in a sample of student work. These discussions provide an opportunity for students to reflect on mathematical thinking of others and will help students avoid making the mistakes themselves.

DIFFERENTIATE INSTRUCTION

For the 10.4.7 Wrap-Up, students are asked to write an explanation of the similarities and differences between solving inequalities with positive and negative coefficients. If students struggle expressing themselves in writing, consider allowing them to circle the steps that are similar and underline the steps that are different and use an audio recorder or speech-to-text software.

10.4.1 WARM-UP: BELL WORK

TEACHER MOVES

The Warm-Up reinforces students' skills operating with rational numbers. Students may need a review of how to add, subtract, multiply, or divide with decimals or fractions. Creating and displaying an anchor chart could be beneficial. Consider not allowing students to use a calculator on the Warm-Up.



10.4.1 Warm-Up: Bell Work

- Using substitution, determine whether the given numerical value is part of the solution set of the inequality. Show your work.
 - Is $n = -5.5$ part of the solution set for $-4n > 20$?
 $-4(-5.5) > 20$
 $22 > 20$
Yes, it is a part of the solution set because 22 is greater than 20.
 - Is $y = 8$ part of the solution set for $-6 \leq -\frac{3}{4}y$?
 $-6 \leq -\frac{3}{4}(8)$
 $-6 \leq -6$
Yes, it is a part of the solution set because -6 is equal to -6.
 - Is $b = 2.7$ part of the solution set for $-9.6 > -12.3 + b$?
 $-9.6 > -12.3 + 2.7$
 $-9.6 > -9.6$
No, it is a part of the solution set because -9.6 is equal to -9.6, not greater than.
- Explain how to determine whether a value is part of the solution set for an inequality.
Plug the value into the equation and determine if the inequality is true with the value plugged in. If the inequality is true, then the value is part of the solution set.



10.4.2 Collaboration: The Effect of a Negative Coefficient

1. Two simple inequalities are shown.

$$1 > 0 \text{ and } -1 > 0$$

- Circle the inequality that is true and cross out the inequality that is false.
 - Complete the statement.
If 1 is greater than 0, then the opposite of 1 is less than 0.
 - Write the correct inequalities by filling in a comparison symbol on each line.
 $0 < 1$ and $0 > -1$
 - Verify your answers with your partner on parts A-C and, if necessary, resolve any differences.
2. Work with your partner to complete the following.
- Complete the inequalities in the table based on your understanding of opposites.

Given Inequality	Equivalent Inequality
$0 < -q$	$0 > q$
$0 > -m$	$0 < m$
$-w \leq 0$	$w \geq 0$

In each equivalent inequality, the sign of the variable changed to its opposite and the value 0 remained the same.

- Describe which operation(s) could be applied to both sides of the given inequalities above to result in the signs of the variables changing and the value 0 remaining the same.
Multiplication or division by -1
- Describe what happens to the comparison symbol in the inequality when either of these operations are applied.
The sign flips or changes to the opposite direction.

10.4.2 COLLABORATION: THE EFFECT OF A NEGATIVE COEFFICIENT

DIFFERENTIATE INSTRUCTION

Prompt students to practice reading inequalities from right to left and left to right. This practice will help students who have a hard time reading or interpreting inequalities with the variables on the right.

TEACHER MOVES

Think-Pair-Share Use this routine for question 1d. Ask students to justify their answers to their partners. These discussions will build student confidence and understanding.

ENRICHMENT

Why is it possible that both multiplying by -1 to both sides and dividing by -1 on both sides results in the same solution?



10.4.3 Bring It Together: The Effect of a Negative Coefficient

1. Complete the statements using words from the bank shown. Some words may not be used.

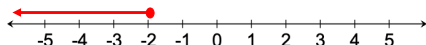
adding	subtracting	multiplying
dividing		opposite

When the variable term in an inequality is negative, the inequality can be rewritten by multiplying or dividing the inequality by -1. In the resulting inequality, each term and the comparison symbol will change to its opposite.

2. Use the inequality $-n \geq 2$ to complete the following.
- Complete the inequality to create an equivalent inequality.

$$n \leq -2$$

- Graph the solution set for $-n \geq 2$.



10.4.3 BRING IT TOGETHER: THE EFFECT OF A NEGATIVE COEFFICIENT

DIFFERENTIATE INSTRUCTION

ELL students may benefit from visual representations for each word in the word bank.

ENRICHMENT

Which inequality, $-n \geq 2$ or $n \leq -2$, is easier to graph on a number line?

10.4.4 GUIDED INSTRUCTION: SOLVING ONE-STEP INEQUALITIES WITH MULTIPLICATION AND DIVISION

TEACHER MOVES

Note that c cannot equal 0 for the multiplication property of inequality. That is why the property states c is either positive or negative.

Ask students to discuss with their partners why c cannot equal 0. (For the multiplication property, c cannot equal 0 because the two expressions would be equal, or both be 0, so the inequality statement would no longer be true.)

DIFFERENTIATE INSTRUCTION

Consider initially presenting the properties using concrete examples (with numbers). After understanding the concrete examples, students will be ready for the abstract generalizations presented with variables.

TEACHER MOVES

Note that c cannot equal 0 for the division property of inequality. That is why the property states that c is either positive or negative.

Ask students to discuss with their partners why c cannot equal 0. (Since division by 0 is undefined, it is not possible to divide either expression by 0.)

If students forget to reverse the direction of the inequality symbol when multiplying by -5 , prompt them to check their work by substituting in a value for c . This approach will empower students to self-regulate as they identify and correct their own mistakes.



10.4.4 Guided Instruction: Solving One-Step Inequalities with Multiplication and Division

As with solving one-step inequalities with addition and subtraction, properties of inequality can be used to solve inequalities with multiplication and division. As discovered in the Collaboration, though, when the value that is being multiplied or divided is negative, this changes both the values in the inequality *and* the comparison symbol to their opposites.

Multiplying both sides of an inequality by the same value can be used to solve an inequality by the **multiplication property of inequality**.



Multiplication Property of Inequality

If $a > b$ and $c > 0$, then $a \times c > b \times c$.

If $a > b$ and $c < 0$, then $a \times c < b \times c$.

- Find the solution to both inequalities using the multiplication property of inequality.

$\frac{y}{4} \leq 2.1$ $4 \times \frac{y}{4} \leq 2.1 \times 4$ $y \leq 8.4$	$\frac{y}{-4} \leq 2.1$ $-4 \times \frac{y}{-4} \leq 2.1 \times -4$ $y \geq -8.4$
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Dividing both sides of an inequality by the same value can be used to solve an inequality by the **division property of inequality**.



Division Property of Inequality

If $a > b$ and $c > 0$, then $a \div c > b \div c$.

If $a > b$ and $c < 0$, then $a \div c < b \div c$.

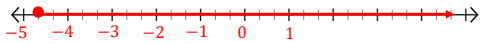
- Find the solution to both inequalities using the division property of inequality.

$-25 > 5c$ $\frac{-25}{5} > \frac{5c}{5}$ $-5 > c$	$-25 > -5c$ $\frac{-25}{-5} > \frac{-5c}{-5}$ $5 < c$
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10.4.5 Guided Practice: Solving One-Step Inequalities with Multiplication and Division

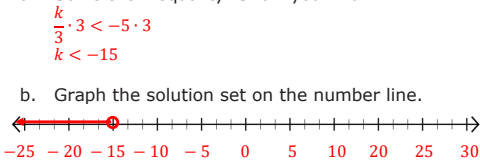
- Use the inequality $7 \geq -\frac{3}{2}y$ to complete the following.
 - Solve the inequality. Show your work.
 - Graph the solution set on the number line.



- Use substitution to test a value or values to verify that the solution and the graph are correct.

$$\begin{aligned}
 7 &\geq -\frac{3}{2}(0) \\
 7 &\geq 0 \\
 \text{True}
 \end{aligned}$$

- Use the inequality $\frac{k}{3} < -5$ to complete the following.
 - Solve the inequality. Show your work.
 - Graph the solution set on the number line.



- Use substitution to test a value or values to verify that the solution and the graph are correct.

$$\begin{aligned}
 \frac{k}{3} &< -5 \\
 \frac{-27}{3} &< -5 \\
 -9 &< -5
 \end{aligned}$$

- Gabriel and Lauren made errors when solving the inequality $-4x \leq 12$. Their work is shown.

Gabriel's Work	Lauren's Work
$ \begin{array}{r} -4x \leq 12 \\ +4 \quad +4 \\ \hline x \leq 16 \end{array} $	$ \begin{array}{r} -4x \leq 12 \\ -4 \quad -4 \\ \hline x \leq -3 \end{array} $

- Explain which mistake(s) each student made.

Gabriel's Error	Lauren's Error
Gabriel added 4 instead of dividing by -4.	Lauren did not flip the sign to be the opposite direction.

- Find the correct solution to $-4x \leq 12$.

$$\begin{aligned}
 -4x &\leq 12 \\
 \frac{-4}{-4}x &\geq \frac{12}{-4} \\
 x &\geq -3
 \end{aligned}$$

10.4.5 GUIDED PRACTICE: SOLVING ONE-STEP INEQUALITIES WITH MULTIPLICATION AND DIVISION

TEACHER MOVES

Discuss how and why multiplying by the reciprocal in question 1a works. Students may benefit from being reminded about the multiplicative inverse property.

TEACHER MOVES

Students may flip the inequality sign because they see there is a negative in the problem. As you circulate, listen for this misconception and address it with the student.

TEACHER MOVES

Students discuss errors they identified and corrected in a sample of student work. These discussions provide an opportunity for students to reflect on the mathematical thinking of others and will help students avoid making the mistakes themselves.

10.4.6 YOUR TURN!

QUESTIONING

Consider extending student learning by having them determine how many days Frankie would need to work to earn the gaming system. Students can also write and solve their own inequalities based on an item they wish to purchase and the rate of pay they earn.



10.4.6 Your Turn!

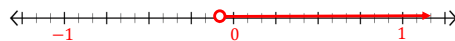
- Frankie got a summer job as a lifeguard at the local pool, where he will earn \$9.50 per hour. He wants to make at least \$299 to buy the newest gaming system. The inequality $299 \leq 9.50x$ mathematically represents this situation where x is the number of hours Frankie works. Find the solution to the inequality to determine the minimum number of hours Frankie must work to save enough money for the gaming system. Round your answer to the nearest hundredth.

$$\begin{aligned} 299 &\leq 9.50x \\ \frac{299}{9.50} &\leq \frac{9.50}{9.50}x \end{aligned}$$

$$31.47 \leq x$$

- Solve each inequality. Then, graph its solution set.

$$\begin{aligned} \text{a. } 9x &> -\frac{3}{4} \\ 9x &> -\frac{3}{4} \\ x &> -\frac{3}{4} \times \frac{1}{9} \\ x &> -\frac{1}{12} \end{aligned}$$

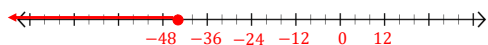


DIFFERENTIATE INSTRUCTION

Consider displaying an anchor chart with instructions, examples, and visuals for solving one-step inequalities with multiplication and division.

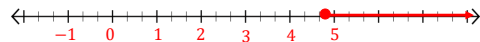
$$\text{b. } 5.5 \leq -\frac{n}{8}$$

$$\begin{aligned} -8 \times 5.5 &\leq -\frac{n}{8} \times -8 \\ -44 &\geq n \end{aligned}$$



$$\text{c. } -3.6y \leq -17.1$$

$$\begin{aligned} -3.6y &\leq -17.1 \\ \frac{-3.6}{-3.6}y &\geq \frac{-17.1}{-3.6} \\ y &\geq 4.75 \end{aligned}$$





10.4.7 Wrap-Up: Cool Down

Two inequalities are shown.

$2.5x \geq -10$ $-2.5x \geq -10$

1. Complete the table to describe how the process of solving each inequality is similar and different.

How is the process of solving each inequality similar?	How is the process of solving each inequality different?
<i>You must divide by the coefficient on both sides.</i>	<i>When dividing by the coefficient in the second equation the inequality sign must change to the opposite direction.</i>

10.4.7 WRAP-UP: COOL DOWN

DIFFERENTIATE INSTRUCTION

If students struggle expressing themselves in writing, consider allowing them to circle the steps that are similar and underline the steps that are different and use an audio recorder or speech-to-text software.